Sum Rate Maximization for RSMA-Assisted CF mMIMO Networks with SWIPT Users

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***Abstract*—This letter develops a rate-splitting multiple access (RSMA)-assisted cell-free (CF) massive multiple input multiple output network comprising simultaneous wireless information and power transfer (SWIPT) users. Specifically, access points (APs) utilize RSMA to serve all users concurrently while minimizing inter-user interference and delivering more power to SWIPT users for energy harvesting. We develop optimal common and private beamforming at the APs and optimal power-splitting factors at users to maximize the user sum rate, subject to the users' minimum rate and EH requirements. In comparison to the conventional CF SWIPT benchmark, the suggested system with 36 APs and 3 users delivers $28.2\,\%$ of sum rate gain.**

***Index Terms*—Cell-free communication, Rate-splitting multiple access, Simultaneous wireless information and power transfer.**

I. Introduction

Emerging 6G wireless networks, aiming for superior data throughput, latency, connectivity, reliability, and energy efficiency (EE), may rely on cell-free (CF) massive multiple input multiple output (mMIMO), rate-splitting multiple access (RSMA), and simultaneous wireless information and power transfer (SWIPT) [1]. Although CF mMIMO offers improved spatial efficiency (SE) and EE, expands coverage, and enhances service quality uniformly for all users, it needs higher front-haul/back-haul requirements to support distributed access points (APs) [2]. These APs connect to a central processing unit (CPU) and serve multiple users concurrently via the front-haul link [2]. RSMA segregates user messages into common and private components [3]. The common parts are jointly encoded into a single stream, while the private parts are encoded separately [3]. Employing successive interference cancellation (SIC), users first decode the common stream to eliminate it before decoding their private stream, allowing partial interference decoding [3]. RSMA exhibits superior SE and EE compared to traditional solutions like SDMA and non-orthogonal multiple access [3]. Finally, SWIPT serves as an alternative power source, effectively extending battery life for energy-constrained and low-power wireless nodes [4].

A. Motivation and Our Contributions

The performance gains of RSMA-assisted CF mMIMO with SWIPT users (Fig. 1) have never been quantified before. This work is the first to do so. In our system model, the APs utilize RSMA to concurrently transmit downlink user signals while managing inter-user interference and providing sufficient power to SWIPT users for energy harvesting (EH). To boost harvested power and achieved rates, we maximize the user sum rate by jointly optimizing the APs' common and private beamforming and users' power-splitting (PS) factors while ensuring minimal rate and user's EH requirements. The main contributions are summarized as follows:

- We introduce an innovative RSMA-assisted CF mMIMO network with SWIPT users and maximize the users' sum rate. Our work is the first to outline this problem and provide an extensive solution.
- As the proposed optimization problem, $\mathbf{P}$ (10), is non-convex and NP-hard, we develop an optimization algorithm based on alternative optimization (AO) [5], utilizing the semi-definite relaxation (SDR) and fractional programming (FP). Our algorithm incorporates minimum rate and power targets for the users, optimizes the APs' common and private beamforming and the PS factors at the users, and mitigates inter-user interference.
- Numerical results demonstrate substantial rate and EH gains. For example, with 36 APs and 3 users, we improve the sum rate by $28.2\,\%$ over the conventional CF mMIMO with optimal beamforming and the received power by $18.3\,\%$ over maximum ratio transmission (MRT).

B. Previous Contributions

RSMA-assisted CF mMIMO networks with SWIPT users are yet to be investigated and present various challenges, including devising beamforming techniques, ensuring user rate demands, enabling efficient EH for SWIPT users via PS determinations, and minimizing interference between users. Addressing these obstacles and achieving optimal performance requires innovative solutions and interventions.

Although prior studies such as [3], [6]–[8] examine RSMA+CF mMIMO without SWIPT, they do not investigate these specific challenges. For instance, [3] focuses on power control, while both [6], [7] explore clustered CF mMIMO with RSMA and propose power control strategies and clustering algorithms. Additionally, [8] addresses asynchronous reception in RSMA CF mMIMO. No prior study has ever explored the optimal beamforming design in RSMA-assisted CF mMIMO, and [3], [6]–[8] solely rely on the classical beamforming techniques, such as MRT, zero-forcing, and minimum mean-squared error. In addition, [9] designs an artificial-noise-aided transmit for a multi-user system with a co-located transmitter to provide confidential service, multicast service, and EH for three different types of users. However, none of the work covers the mentioned technical challenges. This letter aims to explore RSMA+CF mMIMO+SWIPT networks, addressing these unexplored challenges.

Notation: Lower-case bold and upper-case bold denote vectors and matrices. A^*, $\mathbf{A}^{\mathrm{T}}$, $\mathbf{A}^{\mathrm{H}}$, $\mathrm{Tr}(\mathbf{A})$, and $\mathrm{diag}(\mathbf{A})$ denote conjugate, transpose, Hermitian transpose, trace, and diagonal operator. $\mathcal{CN}(\boldsymbol{\mu}, \mathbf{R})$ is a complex Gaussian vector

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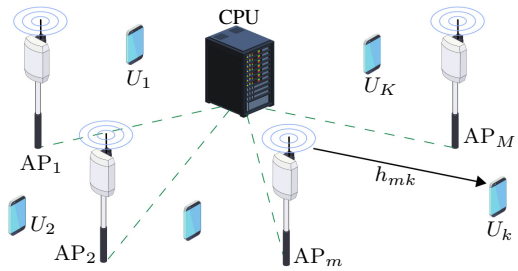

Fig. 1: An RSMA-assisted CF mMIMO system.

with mean $\boldsymbol{\mu}$ and co-variance $\mathbf{R}$. Finally, $\mathcal{K} \triangleq \{1, \ldots, K\}$ and $\mathcal{K}_k \triangleq \mathcal{K} \setminus \{k\}$.

II. System Model and Preliminaries

A. System and Channel Models

Our system comprises $M \geq 1$ APs and $K \geq 1$ SWIPT users, designated as $U_k (k \in \mathcal{K})$ (Fig. 1). SWIPT users utilize PS to divide received power for information decoding and EH purposes [4]. An infinite capacity front-haul/back-haul link interconnects all APs to a CPU. It thus allows APs to simultaneously serve all users within an area, resulting in reduced interference at the users and low computational demand at the APs [2]. This link enables APs to share essential channel state information (CSI). With time division-duplexing mode for both channel estimation and data transmission, CSI can be estimated by employing orthogonal pilots [2], which is highly accurate. Thus, we assume perfect CSI availability to study the critical optimization of the proposed system.

We consider block flat-fading channel models. Within each fading block, the channel response vector between the m-th AP and the U_k, denoted as h_{mk}, is defined as $h_{mk} = \zeta_{mk}^{1/2} \tilde{h}_{mk}$, where $\tilde{h}_{mk} \sim \mathcal{CN}(0, 1)$ captures independent quasi-static Rayleigh fading that remains constant throughout the coherence interval [2]. The term ζ_{mk} represents path-loss/shadowing and is commonly assumed to be constant across multiple coherence intervals [2]. As these long-term channel statistics change slowly, estimation once per tens or hundreds of coherence intervals is sufficient [2].

B. Transmission Model

The APs employ RSMA to provide access to the users. This means the message d_k of U_k is split into a common part $d_{c,k}$ and a private part $d_{p,k}$, i.e., $d_k = \{d_{c,k}, d_{p,k}\}$ for $k \in \mathcal{K}$. The common parts of all users, i.e., $\{d_{c,1}, \ldots, d_{c,K}\}$ are jointly encoded into the common stream s_c, while the private parts, i.e., $\{d_{p,1}, \ldots, d_{p,K}\}$ are respectively encoded into private streams $\{s_1, \ldots, s_K\}$, which are assumed to be independent [3]. The m-th AP linearly precodes the common stream and the private streams before transmission using the precoders $w_{mc} \in \mathbb{C}$ and $\{w_{mi}\}_{i \in \mathcal{K}} \in \mathbb{C}$, respectively. The m-th AP transmitted signal can be expressed as $x_m = w_{mc} s_c + \sum_{i \in \mathcal{K}} w_{mi} s_i$. Here, s_c and $\{s_i\}_{i \in \mathcal{K}}$ are independent of each other [3].

1) Users' Received Data & EH: U_k receives signals from all APs. Assuming the propagation delay differences for all signals are negligible [2], U_k's received signal is given as

$$y'_k = \sum\nolimits_{m \in \mathcal{M}} h^*_{mk} x_m + z_k \tag{1}$$

$$= \sum\nolimits_{m \in \mathcal{M}} h^*_{mk} w_{mc} s_c + \sum\nolimits_{m \in \mathcal{M}} \sum\nolimits_{i \in \mathcal{K}} h^*_{mk} w_{mi} s_i + z'_k,$$

where z_k is the additive white Gaussian noise (AWGN) at U_k user having zero mean and variance of σ_z^2 such that $z'_k \sim \mathcal{CN}(0, \sigma_z^2)$. Defining $\mathbf{h}_k \triangleq [h_{1k}, \ldots, h_{Mk}]^{\mathrm{T}} \in \mathbb{C}^{M \times 1}$, $\mathbf{w}_c \triangleq [w_{1c}, \ldots, w_{Mc}]^{\mathrm{T}} \in \mathbb{C}^{M \times 1}$, and $\mathbf{w}_k \triangleq [w_{1k}, \ldots, w_{Mk}]^{\mathrm{T}} \in \mathbb{C}^{M \times 1}$, the received signal at U_k (1) is rearranged as follows:

$$y'_k = \mathbf{h}_k^{\mathrm{H}} \mathbf{w}_c s_c + \sum\nolimits_{i \in \mathcal{K}} \mathbf{h}_k^{\mathrm{H}} \mathbf{w}_i s_i + z'_k. \tag{2}$$

U_k splits the received signal (2) into two parts based on the PS factor, $\theta_k \in (0, 1)$; (i) θ_k portion for information decoding and (ii) the rest, i.e., $(1 - \theta_k)$, for EH. The input for the information decoding module is thus given as

$$y_k = \sqrt{\theta_k} \mathbf{h}_k^{\mathrm{H}} \mathbf{w}_c s_c + \sqrt{\theta_k} \sum\nolimits_{i \in \mathcal{K}} \mathbf{h}_k^{\mathrm{H}} \mathbf{w}_i s_i + z_k, \tag{3}$$

where $z_k \sim \mathcal{CN}(0, \sigma^2)$ is the AWGN at the information decoding module. The EH module of U_k converts its input power into direct current. The input power to U_k's EH module can be given as $(1 - \theta_k) p_k^{\mathrm{in}}$, where

$$p_k^{\mathrm{in}} = |\mathbf{h}_k^{\mathrm{H}} \mathbf{w}_c|^2 + \sum\nolimits_{i \in \mathcal{K}} |\mathbf{h}_k^{\mathrm{H}} \mathbf{w}_i|^2. \tag{4}$$

2) EH Model at Users: The harvested power, p_k^{h} for $k \in \mathcal{K}$, can be modeled as a linear or nonlinear function of the input power, $(1 - \theta_k) p_k^{\mathrm{in}}$. Although the linear model is widely used due to its simplicity, it ignores saturation and sensitivity characteristics of actual EH circuits [10]. Consequently, a parametric nonlinear EH model based on the sigmoid function is popular [10]. It suggests $p_k^{\mathrm{h}} = \Phi((1 - \theta_k) p_k^{\mathrm{in}})$, where $\Phi(\cdot)$ is give in [10, Eqn. (4)], which is utilized in our study. However, our approach can make use of any other model as well.

Remark 1. *Regardless of the EH model, the activation threshold (p_b) of a user EH circuit, i.e., the minimum power necessary to wake up the EH circuit, is a critical parameter. This arises due to the diode's turn-on voltage characteristics [10]. Therefore, the main goal of our problem formulation is to satisfy this critical requirement for each user. In particular, our problem formulation ensures that $p_k^{\mathrm{h}} \geq p_b$, which eventually satisfies the activation of the EH circuit [10]. As this function is monotonically increasing [10], this requirement translates into $(1 - \theta_k) p_k^{\mathrm{in}} \geq p'_b$, where $p'_b \triangleq \Phi^{-1}(p_b)$ and $\Phi^{-1}(p_b) = b_{\mathrm{N}} - \frac{1}{a_{\mathrm{N}}} \ln\left(\frac{M_{\mathrm{N}} - p_b}{p_b}\right)$. Here, $\Phi^{-1}(p_b)$ is a constant. As a result, this reformulation can handle any linear or nonlinear EH model, irrespective of their properties. Other nonlinear models may be found in [4], [11].*

C. Achievable Rates

In this section, we derive the achievable common and private rates of the users. To decode the common message, s_c, U_k treats the private signals as interference. Thus, by treating the second term in (3) as additive noise, the common data rate at U_k is derived as

$$\mathcal{R}_{c,k} = \log_2(1 + \gamma_{c,k}), \tag{5}$$

where the received SINR at U_k for decoding s_c is given as

$$\gamma_{c,k} = \frac{\theta_k |\mathbf{h}_k^{\mathrm{H}} \mathbf{w}_c|^2}{\theta_k \sum_{i \in \mathcal{K}} |\mathbf{h}_k^{\mathrm{H}} \mathbf{w}_i|^2 + \sigma^2}. \tag{6}$$

To guarantee that all users successfully decode the common stream, the rate of decoding the common stream s_c should not exceed $\mathcal{R}_c = \min_{k\in\mathcal{K}}\mathcal{R}_{c,k}$ [12]. As $\mathcal{R}_c$ is shared by K users, we have $\sum_{k\in\mathcal{K}} C_k = \mathcal{R}_c$, where C_k is the portion of the common rate at U_k allocated for $d_{c,k}$.

Next, U_k subtracts the estimate $\hat{s}_c$ from the received signal and decodes the desired private message, s_k. However, due to the limited dynamic range of the receivers, the common signal cannot be perfectly removed even with the perfect CSI [12]. It is thus assumed that the common signal at U_k is partially eliminated before decoding the private data. The private rate of U_k is thus given by

$$\mathcal{R}_{p,k} = \log_2(1+\gamma_{p,k}), \tag{7}$$

where the received SINR at U_k for decoding s_k is

$$\gamma_{p,k} = \frac{\theta_k|\mathbf{h}_k^{\mathrm{H}}\mathbf{w}_k|^2}{\delta_c\theta_k|\mathbf{h}_k^{\mathrm{H}}\mathbf{w}_c|^2 + \theta_k\sum_{i\in\mathcal{K}_k}|\mathbf{h}_k^{\mathrm{H}}\mathbf{w}_i|^2+\sigma^2}, \tag{8}$$

where δ_c accounts for SIC qualities for removing the common signal. The total achievable rate of U_k, including the portion of the common rate and the private rate is expressed as [12]

$$\mathcal{R}_k = C_k + \mathcal{R}_{p,k}. \tag{9}$$

III. Problem Formulation

Herein, we optimize the performance of the proposed RSMA-assisted CF mMIMO setup with SWIPT users. In particular, we focus on maximizing the users' sum rate by cohesively optimizing the APs' transmit beamforming, $\mathbf{w}_c$ and $\{\mathbf{w}_k\}_{k\in\mathcal{K}}$, common rates at the users, $\{C_k\}_{k\in\mathcal{K}}$, and the users' PS factors, $\{\theta_k\}_{k\in\mathcal{K}}$. We denote the set of these optimization variables as $\mathcal{A} = \{\mathbf{w}_c, \{\mathbf{w}_k\}_{k\in\mathcal{K}}, \{C_k\}_{k\in\mathcal{K}}, \{\theta_k\}_{k\in\mathcal{K}}\}$. The optimization problem is thus formulated as follows:

$$\begin{aligned}
\mathbf{P}:\ & \max_{\mathcal{A}} \sum\nolimits_{k\in\mathcal{K}} \mathcal{R}_k, && \text{(10a)}\\
\text{s.t}\ & \mathcal{R}_{p,k} \geq \mathcal{R}_{p,k}^{\mathrm{th}}, \quad \forall k, && \text{(10b)}\\
& \sum\nolimits_{k\in\mathcal{K}} C_k \leq \mathcal{R}_c, && \text{(10c)}\\
& (1-\theta_k)p_k^{\mathrm{in}} \geq \Phi^{-1}(p_b), \quad \forall k, && \text{(10d)}\\
& |w_{m,c}|^2 + \sum\nolimits_{i\in\mathcal{K}}|w_{mi}|^2 \leq p_{\max}, \quad \forall m, && \text{(10e)}\\
& 0 \leq C_k, \quad \forall k, && \text{(10f)}\\
& 0 < \theta_k < 1, \quad \forall k, && \text{(10g)}
\end{aligned}$$

where (10b) sets the minimum rate targets for users to decode the private stream. Constraints (10c) and (10f) guarantee the successful decoding of the common stream, while (10d) ensures the minimum power requirements at the users for EH. Besides, (10e) is the per AP transmit power constraint with maximum allowable transmit power $p_{\max}$. Finally, (10g) is the PS factor at the users.

IV. Proposed Solutions

Problem $\mathbf{P}$ is non-convex due to the non-convex objective function and constraints involving products of variables. Hence, we utilize the AO technique, splitting the problem into two sub-problems and addressing them using SDR and FP.

A. Transmit Beamforming and Common Rate

For given $\{\theta_k\}_{k\in\mathcal{K}}$, $\mathbf{P}$ becomes a transmit beamforming and common rate optimization problem. We first define the matrices $\mathbf{W}_c \triangleq \mathbf{w}_c\mathbf{w}_c^{\mathrm{H}}$, and $\mathbf{W}_k \triangleq \mathbf{w}_k\mathbf{w}_k^{\mathrm{H}}$, where $\mathbf{W}_c$ and $\mathbf{W}_l$ are semi-definite matrices with rank one constraints, i.e., $\mathrm{Rank}(\mathbf{W}_c) = 1$ and $\mathrm{Rank}(\mathbf{W}_k) = 1$. By utilizing SDR techniques to relax the highly non-convex rank one constraints, the resultant problem can be formulated as follows:

$$\begin{aligned}
\mathbf{P}_{\mathrm{w}}:\ & \max_{\mathbf{W}_c,\mathbf{W}_k,C_k} \sum\nolimits_{k\in\mathcal{K}} C_k + \digamma_k, && \text{(11a)}\\
\text{s.t}\ & \Big(\theta_k\Big(\delta_c\mathrm{Tr}(\mathbf{H}_k\mathbf{W}_c) + \sum\nolimits_{i\in\mathcal{K}_k}\mathrm{Tr}(\mathbf{H}_k\mathbf{W}_i)\Big)+\sigma^2\Big)\\
& -\theta_k\mathrm{Tr}(\mathbf{H}_k\mathbf{W}_k)/\gamma_{p,k}^{\mathrm{th}} \leq 0, && \text{(11b)}\\
& \sum\nolimits_{k\in\mathcal{K}} C_k \leq \min(\Lambda_k), && \text{(11c)}\\
& \mathrm{Tr}(\mathbf{H}_k\mathbf{W}_c) + \sum\nolimits_{i\in\mathcal{K}}\mathrm{Tr}(\mathbf{H}_k\mathbf{W}_i) \geq p_{\mathrm{th}}/(1-\theta_k), && \text{(11d)}\\
& \mathrm{diag}(\mathbf{W}_c) + \sum\nolimits_{i\in\mathcal{K}}\mathrm{diag}(\mathbf{W}_i) \leq p_{\max}\mathbf{1}_M, && \text{(11e)}\\
& 0 \leq C_k, && \text{(11f)}
\end{aligned}$$

where $\mathbf{H}_k = \mathbf{h}_k\mathbf{h}_k^{\mathrm{H}}$, $p_{\mathrm{th}} = \Phi^{-1}(p_b)$, $\gamma_{p,k}^{\mathrm{th}} \triangleq 2^{\mathcal{R}_{p,k}^{\mathrm{th}}} - 1$, and $\mathbf{1}_M \triangleq [1,\ldots,1]^{\mathrm{T}} \in \mathbb{R}^{M\times 1}$. Since user rate is a monotonically increasing function of its argument, i.e., SINRs, we replace $\mathcal{R}_{p,k}$ with $\gamma_{p,k}$ to obtain (11b). As (10a) and (10c) are not convex functions of the optimization variables, we utilize the successive convex approximation (SCA) method to linearize the objective and constraint as given in (11a) and (11c), where $\digamma_k$ and Λ_k are defined in (12) and (13), respectively, with $A_k \triangleq \theta_k\Big(\delta_c\mathrm{Tr}(\mathbf{H}_k\mathbf{W}_c^{(t)}) + \sum_{i\in\mathcal{K}_k}\mathrm{Tr}(\mathbf{H}_k\mathbf{W}_i^{(t)})\Big)+\sigma^2$ and $B_k \triangleq \theta_k\sum_{i\in\mathcal{K}}\mathrm{Tr}(\mathbf{H}_k\mathbf{W}_i^{(t)})+\sigma^2$. Here, $(\cdot)^{(t)}$ denotes the previous iteration values of respective variables. Hence, the relaxed $\mathbf{P}_{\mathrm{w}}$ is a standard semi-definite programming (SDP) problem and can be solved using the CVX Matlab tool [13].

If the SDR solution satisfies $\mathrm{Rank}(\mathbf{W}_c) = 1$ and $\mathrm{Rank}(\mathbf{W}_k) = 1$, the optimal transmit beamformers are obtained by eigenvalue decomposition [14]. Let the eigenvalue decomposition of $\mathbf{W}$ (i.e., $\mathbf{W} \in \{\mathbf{W}_c^o, \{\mathbf{W}_k^o\}_{k\in\mathcal{K}}\}$) to be $\mathbf{W} = \mathbf{U}\boldsymbol{\Sigma}\mathbf{U}^H$ where $\mathbf{U}$ is a unitary matrix and $\boldsymbol{\Sigma} = \mathrm{diag}(\lambda_1,\ldots,\lambda_M)$ is a diagonal matrix, both sized $M\times M$. If $\mathbf{W}^o$ is rank one, the optimal transmit beamformer, $\mathbf{w}^o$, is the eigenvector for the maximum eigenvalue. Otherwise, we use Gaussian randomization (GR) to get a near-optimal solution to $\mathbf{P}_{\mathrm{w}}$ [14]. However, satisfying the rank-one constraint by using GR may cause a slight rate degradation $\mathbf{P}_{\mathrm{w}}$.

B. Power Splitting

This sub-problem is associated with optimizing $\{\theta_k\}_{k\in\mathcal{K}}$ for fixed $\{\mathbf{w}_c, \{\mathbf{w}_k\}_{l\in\mathcal{K}}, \{C_k\}_{k\in\mathcal{K}}\}$. As C_k is fixed, we can eliminate it from the objective function. Thereby invoking the FP technique [15], the resultant problem is formulated as

$$\begin{aligned}
\mathbf{P}_\theta:\ & \max_{\theta_k,\lambda_k} \sum\nolimits_{k\in\mathcal{K}} \log_2(1+2\lambda_k\sqrt{Q_k} - \lambda_k^2 V_k), && \text{(14a)}\\
\text{s.t}\ & \gamma_{p,k}^{\mathrm{th}}\Big(\theta_k\Big(\delta_c|\mathbf{h}_k^{\mathrm{H}}\mathbf{w}_c|^2 + \theta_k\sum\nolimits_{i\in\mathcal{K}_k}|\mathbf{h}_k^{\mathrm{H}}\mathbf{w}_i|^2\Big)+\sigma^2\Big)\\
& -\theta_k|\mathbf{h}_k^{\mathrm{H}}\mathbf{w}_k|^2 \leq 0, && \text{(14b)}\\
& \gamma_{c,k}^{\mathrm{th}}\Big(\theta_k\sum\nolimits_{i\in\mathcal{K}}|\mathbf{h}_k^{\mathrm{H}}\mathbf{w}_i|^2+\sigma^2\Big) - \theta_k|\mathbf{h}_k^{\mathrm{H}}\mathbf{w}_c|^2 \leq 0, && \text{(14c)}
\end{aligned}$$

$$\Gamma_k \triangleq \log_2\Big(\theta_k\Big(\delta_c \mathrm{Tr}(\mathbf{H}_k\mathbf{W}_c)+\sum\nolimits_{i\in\mathcal{K}}\mathrm{Tr}(\mathbf{H}_k\mathbf{W}_i)\Big)+\sigma^2\Big)-\log_2(A_k)-\frac{\theta_k\delta_c\mathrm{Tr}(\mathbf{H}_k)}{\ln(2)A_k}\mathrm{Tr}(\mathbf{W}_c-\mathbf{W}_c^{(t)})-\sum\nolimits_{i\in\mathcal{K}_k}\frac{\theta_k\mathrm{Tr}(\mathbf{H}_k)}{\ln(2)A_k}\mathrm{Tr}(\mathbf{W}_i-\mathbf{W}_i^{(t)}) \quad (12)$$

$$\Lambda_k \triangleq \log_2\Big(\theta_k\Big(\mathrm{Tr}(\mathbf{H}_k\mathbf{W}_c)+\sum\nolimits_{i\in\mathcal{K}}\mathrm{Tr}(\mathbf{H}_k\mathbf{W}_i)\Big)+\sigma^2\Big)-\log_2(B_k)-\sum\nolimits_{i\in\mathcal{K}}\frac{\theta_k\mathrm{Tr}(\mathbf{H}_k)}{\ln(2)B_k}\mathrm{Tr}(\mathbf{W}_i-\mathbf{W}_i^{(t)}) \quad (13)$$

$$(1-\theta_k)\Big(|\mathbf{h}_k^{\mathrm{H}}\mathbf{w}_c|^2+\sum\nolimits_{i\in\mathcal{K}}|\mathbf{h}_k^{\mathrm{H}}\mathbf{w}_i|^2\Big)\geq p_{\mathrm{th}}, \quad (14d)$$

$$0<\theta_k<1, \quad (14e)$$

where $\gamma_{c,k}^{\mathrm{th}} \triangleq 2^{\sum_{k\in K}C_k}-1$ and $\{\lambda_k\}_{k\in\mathcal{K}}$ is the auxiliary variable introduced by the FP techniques. Besides, Q_k and V_k are the numerator and denominator of U_k's SINR (8) as functions of θ_k. This reformulated $\mathbf{P}_\theta$ can be considered as a two-part optimization problem, i.e., (i) an outer optimization over θ_k with fixed λ_k and (ii) an inner optimization over λ_k with fixed θ_k. For given $\{\theta_k\}_{k\in\mathcal{K}}$, the inner optimization is thus a convex in $\{\lambda_k\}_{k\in\mathcal{K}}$ with the optimal value of $\lambda_k^o = \sqrt{Q_k}/V_k$ [15]. For given $\{\lambda_k\}_{k\in\mathcal{K}}$, $\mathbf{P}_\theta$ (the outer optimization) is convex and can be addressed via CVX Matlab [13], [15].

Algorithm 1 presents the overall steps to solve the proposed problem $\mathbf{P}$ (10). It begins with the initialization of $\{\theta_k\}_{k\in\mathcal{K}}$ to random feasible values. In each iteration, it refines the values of $\{\mathbf{w}_c, \{\mathbf{w}_k\}_{k\in\mathcal{K}}, \{C_k\}_{k\in\mathcal{K}}\}$ and $\{\theta_k\}_{k\in\mathcal{K}}$ alternatively until the normalized improvement of the sum rate is smaller than $\epsilon = 1\times10^{-3}$.

Remark 2. *The feasibility region of matrix* $\mathbf{P}$ *is primarily governed by the constraints* (10c)*,* (10e)*,* (10f)*, and* (10g)*. To initialize the variables required for running Algorithm 1, we adopt a strategy of generating random initial values from a uniform distribution within the range [0,1] for* θ_k*. If the set of* θ_k *does not meet the feasibility conditions of* $\mathbf{P}_{\mathrm{w}}$*, our algorithm reiterates the initialization process, generating new* θ_k *values from the same uniform distribution. This iterative process persists until suitable initial values for* θ_k *are identified, satisfying the constraints outlined in* (10)*. Such an approach is commonplace in the wireless literature for similar algorithms [14], [15], and we opt for consistency by adopting this methodology. Simulation results corroborate the robustness of our algorithm, demonstrating rapid convergence irrespective of the initial values chosen.*

C. Algorithm Convergence

Algorithm 1, being an AO algorithm, converges as long as the individual optimization blocks converge [5]. We employ the SDR method to jointly solve $\{\mathbf{w}_c, \{\mathbf{w}_k\}_{k\in\mathcal{K}}, \{C_k\}_{k\in\mathcal{K}}\}$, while $\{\theta_k\}_{k\in\mathcal{K}}$ is solved by using FP technique. Both SDR and FP methods are well established with provable convergence [14], [15], ensuring the convergence of Algorithm 1. Additionally, the SCA employed in $\mathbf{P}_{\mathrm{w}}$ to convexify the objective function satisfies the necessary conditions for retaining the original problem's convergence [16]. Nonetheless, our simulation results also corroborate the validity of this claim (Fig. 2).

D. Computational Complexity

The computational complexity of Algorithm 1 mainly depends on the SDR and FP components.

Algorithm 1 : AO Algorithm

1: **Input**: Set the iteration counter $t=0$, the convergence tolerance $\epsilon>0$, initial feasible solution $\{\theta_k\}_{k\in\mathcal{K}}$. Initialize the objective function value $F^{(0)}=0$.
2: **while** $\frac{F^{(t+1)}-F^{(t)}}{F^{(t+1)}}\geq\epsilon$ **do**
3: Solve $\mathbf{P}_{\mathrm{w}}$ (11) to obtain (sub-)optimal transmit beamforming, $\{\mathbf{w}_c^{(t+1)}, \mathbf{w}_k^{(t+1)}\}$, and the common rate, $C_k^{(t+1)}$, by applying GR to recover the rank-one solution.
4: Solve $\mathbf{P}_\theta$ (14) to obtain the optimal PS coefficients, $\theta_k^{(t+1)}$.
5: Calculate the objective function value $F^{(t+1)}$.
6: Set $t\leftarrow t+1$;
7: **end while**
8: **Output**: Optimal solutions $\mathcal{A}^o$.

TABLE I: Simulation parameters.

Parameter	Value	Parameter	Value
f_c	3 GHz	δ_c	-20 dB
B	10 MHz	p_b	-20 dBm
N_f	10 dB	$p_{\max}$	30 dBm
K	3	M_{N}	20×10^{-3} W
$\mathcal{R}_{p,k}^{\mathrm{th}}$	2 bps/Hz	$\{a_{\mathrm{N}}, b_{\mathrm{N}}\}$	{6400, 0.003}

1) Complexity of solving $\{\mathbf{w}_c, \mathbf{w}_k, C_k\}$*:* This SDR subproblem is tackled using SDP, specifically the interior-point method. From [17, Th. 3.12], computational complexity for a SDP problem with m SDP constraints with an $n\times n$ positive semi-definite matrix is given by $\mathcal{O}\left(\sqrt{n}\log(1/\epsilon)\left(mn^3+m^2n^2+m^3\right)\right)$, where $\epsilon>0$ is the solution accuracy. In $\mathbf{P}_{\mathrm{w}}$, $n=M$ and $m=3K+2$, and the approximate computational complexity is $\mathcal{O}\left(KM^3\sqrt{M}\log(1/\epsilon)\right)$.

2) Complexity of solving θ_k*:* Algorithm 1 is a multistage iterative method. The outer loop optimizes $\{\theta_k\}_{k\in\mathcal{K}}$, while the inner loop updates the auxiliary variable, $\{\lambda\}_{k\in\mathcal{K}}$. As per [15], the computational complexity is primarily concentrated in the outer loop, in the order of $\mathcal{O}(K^3M^3)$.

V. Simulation Results

We adopt the 3GPP urban micro (UMi) model to model the large-scale fading ζ_{mk} with $f_c = 3\,\mathrm{GHz}$ operating frequency [18, Table B.1.2.1]. In addition, we model the AWGN variance, σ^2, as $\sigma^2 = 10\log_{10}(N_0BN_f)$ dBm, where $N_0 = -174\,\mathrm{dBm/Hz}$, B is the bandwidth, and N_f is the noise figure. We consider an $80\times80\,\mathrm{m}^2$ square area, where the APs are uniformly distributed and the users are randomly distributed within the area. Unless otherwise specified, Table I summarizes the simulation parameters.

For comparative evaluations, we consider the following benchmarks against the proposed RSMA-assisted CF system:

1) *Conventional CF with optimal beamforming:* The APs use the SDMA scheme to serve the users, i.e., $x_m =$

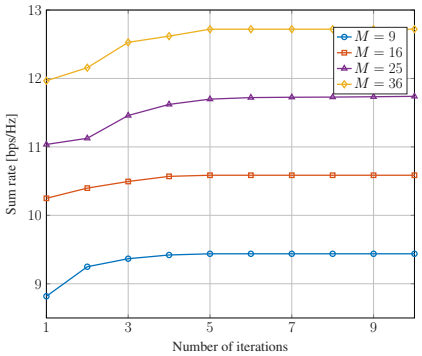

Fig. 2: Convergence of the proposed Algorithm 1.

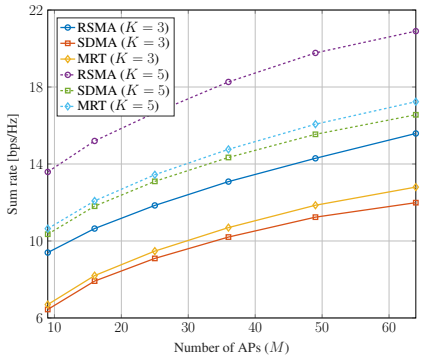

Fig. 3: Sum rate versus the number of APs.

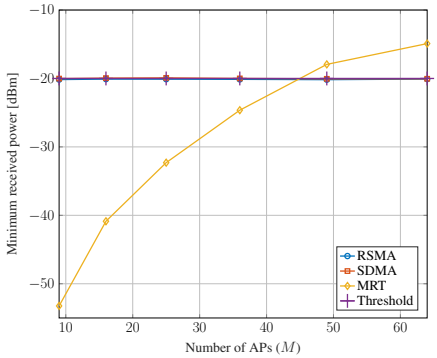

Fig. 4: User received power versus the number of APs.

$\sum_{i\in\mathcal{K}} w_{mi}s_i$. This helps to evaluate the impact of incorporating rate-splitting into CF mMIMO. Our Algorithm 1 handles this as a special case.

2) *Conventional CF with MRT beamforming:* This sets $\mathbf{w}_k = \mathbf{h}_k/\|\mathbf{h}_k\|$ with random PS factors at the users.

Fig. 2 depicts the convergence behavior of Algorithm 1 for $K = 3$. This outputs the APs' transmit beamforming, users' common rates, and the users' PS factors. The stable sum rate achieved by it over several iterations illustrates convergence. To this end, Fig. 2 shows that the sum rate increases rapidly and saturates with the number of iterations. Specifically, regardless of the number of APs, convergence occurs in less than 5 iterations.

Fig.3 examines how the number of APs, M, affects the user sum rate. The RSMA scheme achieves the highest sum rate among SWIPT users. For instance, with 36 APs and 3 users, a $28.2\,\%$ increase is achieved compared to the first benchmark. As inter-user interference decreases, this sum rate continues to improve with an increasing number of users. In contrast, although the MRT scheme outperforms SDMA in terms of sum rate, it exhibits poor EH performance (Fig.4). This deficiency stems from its disregard for the users' EH constraints.

Fig. 4 illustrates the power delivered to the users as a function of the number of APs, M. It is clear that both the proposed RSMA (i.e., Algorithm 1) and SDMA schemes excel at this, satisfying the activation threshold even with a small number of APs. For example, Algorithm 1 scheme delivers a $18.3\,\%$ received power improvement over the MRT scheme with 36 APs. The MRT benchmark, however, requires many more APs to maintain user power targets.

VI. Conclusion

This paper proposed an RSMA-assisted CF mMIMO network with SWIPT users. To maximize user rates and harvested power while efficiently managing inter-user interference, we utilized RSMA at the APs and optimized the common and private beamforming of the APs and the PS factors of the users. We solved the non-convexity of the resulting problem by combining AO, SDR, and FP approaches. Our algorithm maximizes the sum rate while guaranteeing minimal power targets at the users. Our proposed algorithm enables the integration of RSMA, CF mMIMO, and SWIPT, by simultaneously improving harvested power and achieved rates, promoting reciprocal benefits, and improving overall system performance.

References

[1] W. Tong and P. Zhu, *6G: The Next Horizon: From Connected People and Things to Connected Intelligence*. Cambridge University Press, 2021.

[2] Ö. Demir, E. Björnson, and L. Sanguinetti, *Foundations of User-Centric Cell-Free Massive MIMO*, ser. Foundations and trends in signal processing. Now Publishers, 2021.

[3] A. Mishra, Y. Mao, L. Sanguinetti, and B. Clerckx, "Rate-splitting assisted massive machine-type communications in cell-free massive MIMO," *IEEE Commun. Lett.*, vol. 26, no. 6, pp. 1358–1362, Jun. 2022.

[4] D. Galappaththige, R. Shrestha, and G. A. Aruma Baduge, "Exploiting cell-free massive MIMO for enabling simultaneous wireless information and power transfer," *IEEE Trans. Green Commun. Netw.*, vol. 5, no. 3, pp. 1541–1557, Sept. 2021.

[5] J. C. Bezdek and R. J. Hathaway, "Convergence of alternating optimization," *Neural, Parallel & Sci. Comput.*, vol. 11, no. 4, pp. 351–368, Dec. 2003.

[6] A. R. Flores, R. C. de Lamare, and K. V. Mishra, "Clustered cell-free multi-user multiple-antenna systems with rate-splitting: Precoder design and power allocation," *IEEE Trans. Commun.*, vol. 71, no. 10, pp. 5920–5934, Oct. 2023.

[7] M. Sarker and A. O. Fapojuwo, "Uplink power allocation for RSMA-aided user-centric cell-free massive MIMO systems," in *IEEE 97th Veh. Technol. Conf. (VTC2023-Spring)*, Jun. 2023, pp. 1–5.

[8] J. Zheng, J. Zhang, J. Cheng, V. C. M. Leung, D. W. K. Ng, and B. Ai, "Asynchronous cell-free massive MIMO with rate-splitting," *IEEE J. Sel. Areas Commun.*, vol. 41, no. 5, pp. 1366–1382, May 2023.

[9] W. Mei, B. Fu, L. Li, Z. Chen, and C. Huang, "Artificial-noise aided transmit design for multi-user MISO systems with service integration and energy harvesting," in *IEEE Int. Conf. Commun. Workshops (ICC)*, May 2016, pp. 219–225.

[10] E. Boshkovska, D. W. K. Ng, N. Zlatanov, and R. Schober, "Practical non-linear energy harvesting model and resource allocation for SWIPT systems," *IEEE Commun. Lett.*, vol. 19, pp. 2082–2085, Dec. 2015.

[11] Y. Ye, L. Shi, X. Chu, and G. Lu, "Throughput fairness guarantee in wireless powered backscatter communications with HTT," *IEEE Wireless Commun. Lett.*, vol. 10, no. 3, pp. 449–453, Mar. 2021.

[12] C. Xu, B. Clerckx, S. Chen, Y. Mao, and J. Zhang, "Rate-splitting multiple access for multi-antenna joint radar and communications," *IEEE J. Sel. Topics Signal Process.*, vol. 15, no. 6, pp. 1332–1347, Nov. 2021.

[13] S. Boyd and L. Vandenberghe, *Convex Optimization*. Cambridge, U.K.: Cambridge Univ. Press, Mar. 2004.

[14] Q. Wu and R. Zhang, "Intelligent reflecting surface enhanced wireless network via joint active and passive beamforming," *IEEE Trans. Wireless Commun.*, vol. 18, no. 11, pp. 5394–5409, Nov. 2019.

[15] K. Shen and W. Yu, "Fractional programming for communication systems—part i: Power control and beamforming," *IEEE Trans. Signal Process.*, vol. 66, no. 10, pp. 2616–2630, May 2018.

[16] M. Razaviyayn, "Successive convex approximation: Analysis and applications," Ph.D. dissertation, University of Minnesota, Minneapolis, United States, May 2014.

[17] I. P'olik and T. Terlaky, *Interior Point Methods for Nonlinear Optimization*. Berlin, Germany; New York, NY, USA: Springer, 2010.

[18] "3GPP TR 36.814, further advancements for E-UTRA physical layer aspects, V.9.0.0 Rel. 9," Mar. 2010. [Online]. Available: https://portal.3gpp.org/desktopmodules/Specifications/SpecificationDetails.aspx?specificationId=2493